



Digital Pioneers Initiative
DiGiLiANS — Nano Degree Program
Capstone Project Defense — June 2026

Capstone Project: FerroFlux — Intelligent Steel Supply Chain Analytics Platform

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Track: AI — Job Profile: Big Data Engineering

ABSTRACT

- FerroFlux is a real-time Big Data platform for Egyptian steel supply chains, replacing manual spreadsheets with a unified multi-tenant system.
- It uses Kafka for streaming, Spark Medallion ETL, and Airflow for orchestration.
- It enables anomaly detection in pricing, production, and logistics with automated alerts via n8n.
- It provides ML predictions: price forecasting ($R^2 = 0.9578$, MAPE = 1.61%), demand forecasting ($R^2 = 0.8030$, MAPE = 35.74%), and supplier risk classification (94.78% accuracy).
- It reduces decision-making time from days to minutes and supports Industry 4.0 adoption.

Methodology & Approach

- **System Design and Architecture Overview**
FerroFlux is a layered real-time and batch data platform for unified analytics and decision support.
- **Tools, Frameworks, and Technologies Used**
Kafka, Spark, Airflow, FastAPI, Streamlit, PostgreSQL, Docker, and n8n.
- **Dataset, Data Pipeline, or Input Sources**
Market prices, factory telemetry, production data, and shipment tracking processed through an ingestion-to-analytics pipeline.
- **Methods and Algorithms**
Machine learning for price forecasting, demand prediction, and supplier risk scoring, plus anomaly detection with automated alerts.

Results & Key Outcomes

- **Price Forecasting Model**
 $R^2 = 0.9578$ | MAPE = 1.61%
Sub-2% error achieved (target exceeded: $R^2 > 0.90$, MAPE < 5%)
- **Demand Forecasting Model**
 $R^2 = 0.8030$ Strong 30-day directional planning signals (target exceeded: $R^2 > 0.75$)
- **Supplier Risk Classifier**
Accuracy = 94.78% Precision = 100% | Recall = 82.29% (HIGH-risk) Zero false positives in critical risk detection
- **System Reliability**
3 concurrent tenants tested Zero cross-tenant data leakage
- **Performance**
Dashboard load time < 2s Alert dispatch within 15 minutes after threshold breach
- **Usability Testing**
100% task completion rate (5 participants)All core workflows completed without assistance.

Background & Introduction

- Egypt’s steel industry generates large-scale operational and financial data across distributed plants, but many mid-sized producers still rely on spreadsheets, manual reporting, and fragmented ERP systems.
- This creates delays in decision-making and limits the use of data-driven insights for pricing, production, and supplier management.
- FerroFlux addresses this gap by integrating streaming data, batch processing, ML models, and alerting into a unified, real-time analytics platform.

Problem Statement

- Egyptian steel manufacturers face multiple challenges that reduce supply chain efficiency and decision-making quality.
- Price volatility is handled manually without forecasting or anomaly detection.
- Production and demand data are fragmented across factories, preventing unified planning.
- Supplier management is reactive with no risk scoring, increasing disruption risk.
- KPI reporting is retrospective, limiting real-time decision-making.
- Anomalies are detected manually, causing delays in response. Data remains siloed across plants, limiting enterprise-level analytics.



Conclusion & Recommendations

- FerroFlux demonstrates that a focused data engineering system can deliver a production-grade, end-to-end analytics platform for industrial use cases using low-cost, open-source technologies.
- It unifies streaming (Kafka), ETL (Spark Medallion), orchestration (Airflow), ML inference (XGBoost, Random Forest), APIs (FastAPI), dashboards (Streamlit), and alerting (n8n) into a single integrated system.
- The project shows how replacing manual and fragmented workflows with a unified data architecture improves visibility, reduces decision latency, and enables predictive insights across procurement, production, and supplier management.
- Future work includes integrating live steel price feeds, adopting LSTM models for improved forecasting, adding ERP integration, improving demand forecasting with macroeconomic factors, developing a mobile-responsive UI, supporting Arabic language, and enhancing role-based access control.

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